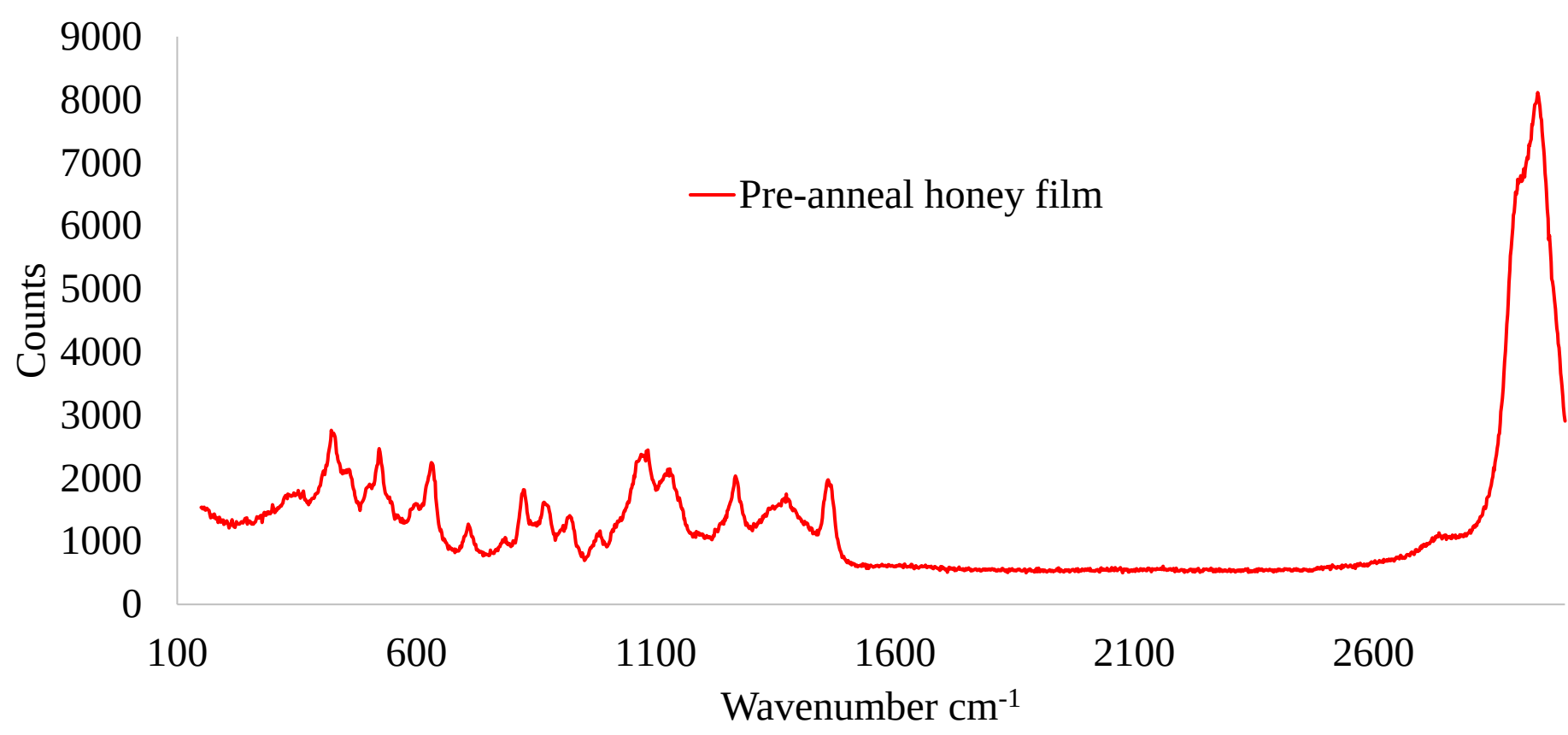


Introduction

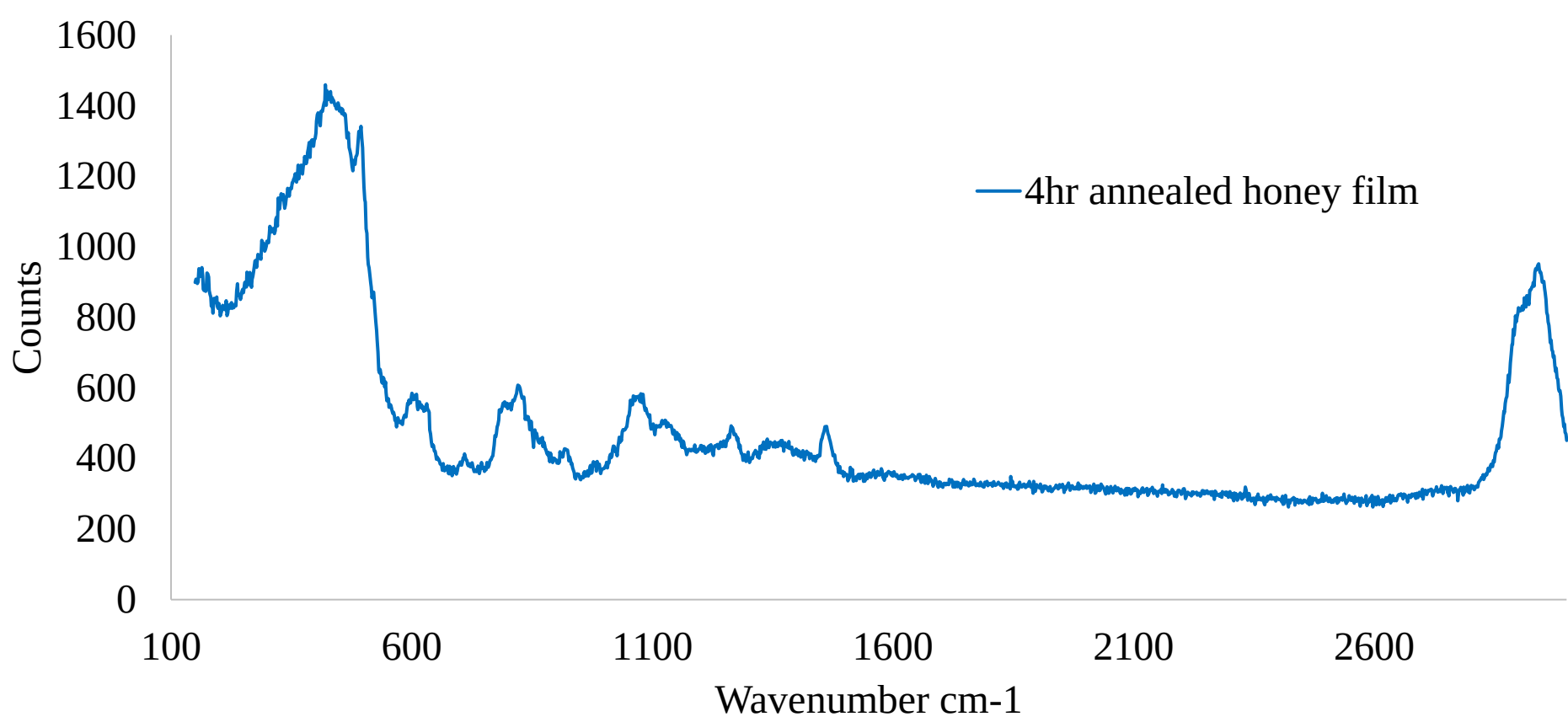
- Honey is being studied lately as a promising biomaterial with potential for use as a sustainable dielectric film in memristor devices. Tuning the honey-CNT film thickness and the concentration of the CNTs will play a vital role in the performance of the devices. While the structural characterization of honey-CNT films is challenging, we demonstrate a strategic combination of several characterization techniques.
- Bright field (BF) optical microscopy shows changes to the honey film when annealed, while dark field (DF) images identify the dispersion of the CNTs.
- Scanning laser microscopy estimates and measures changes to the thickness of the film by annealing and confirms the presence of the CNT clusters.
- Raman spectroscopy identifies characteristic peaks for CNTs, honey and glass. The effect of annealing on the chemical composition of the honey and the thickness of the film are determined.
- Use of low accelerating voltage in SEM enables characterization of CNT clusters embedded in the honey film despite challenges for biomaterials.

Raman Spectroscopy

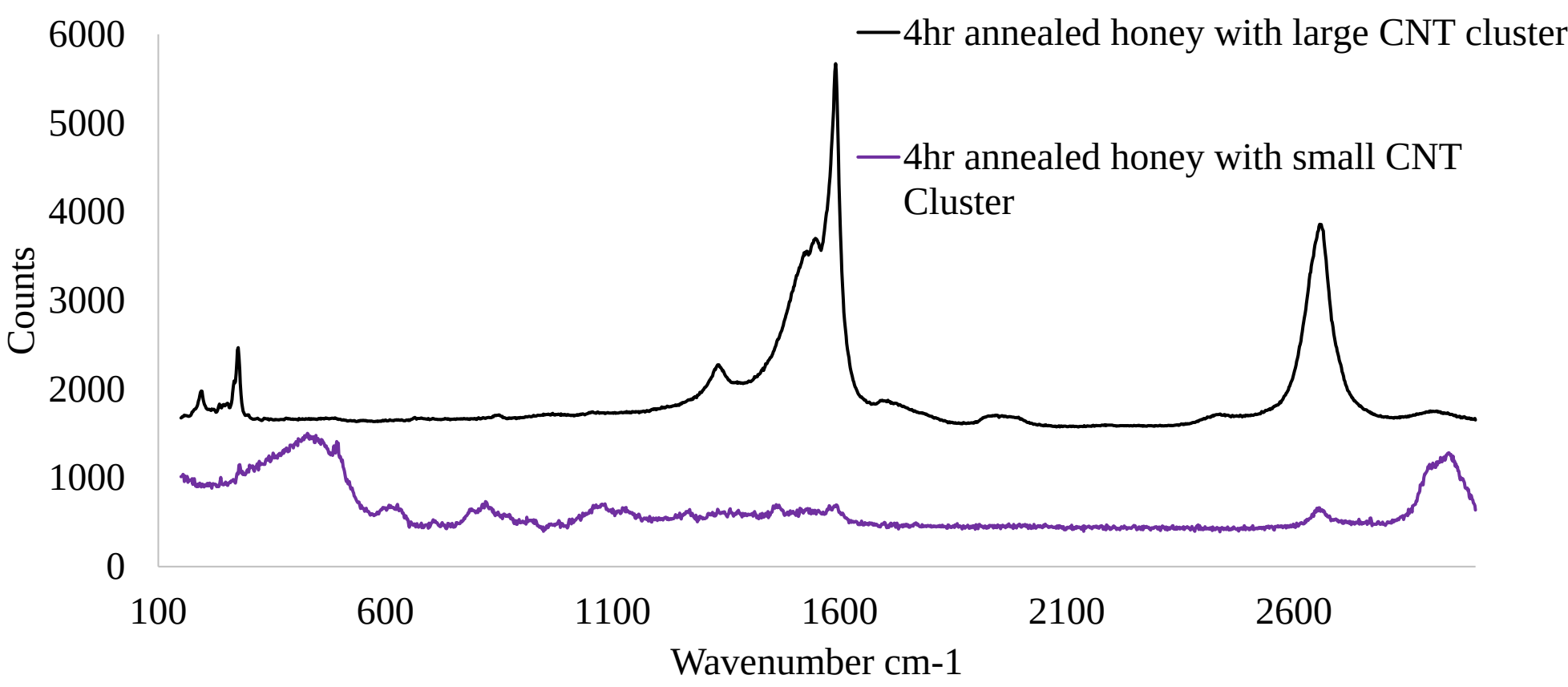
- The CNT clusters, the honey film and the glass substrate are all Raman active and show characteristic peaks.
- By sampling the honey film before and after annealing, the appearance of a peak from the glass substrate demonstrates that the film has become much thinner.
- Furthermore, the size of CNT clusters can qualitatively be identified by the strength of the CNT characteristic peaks. A large CNT cluster only shows characteristic CNT peaks, while a small cluster shows characteristic peaks from all three components (CNT, honey, glass).



The thick pre-annealed honey film shows only characteristic peaks from the honey itself between ~300 cm⁻¹ and ~2900 cm⁻¹



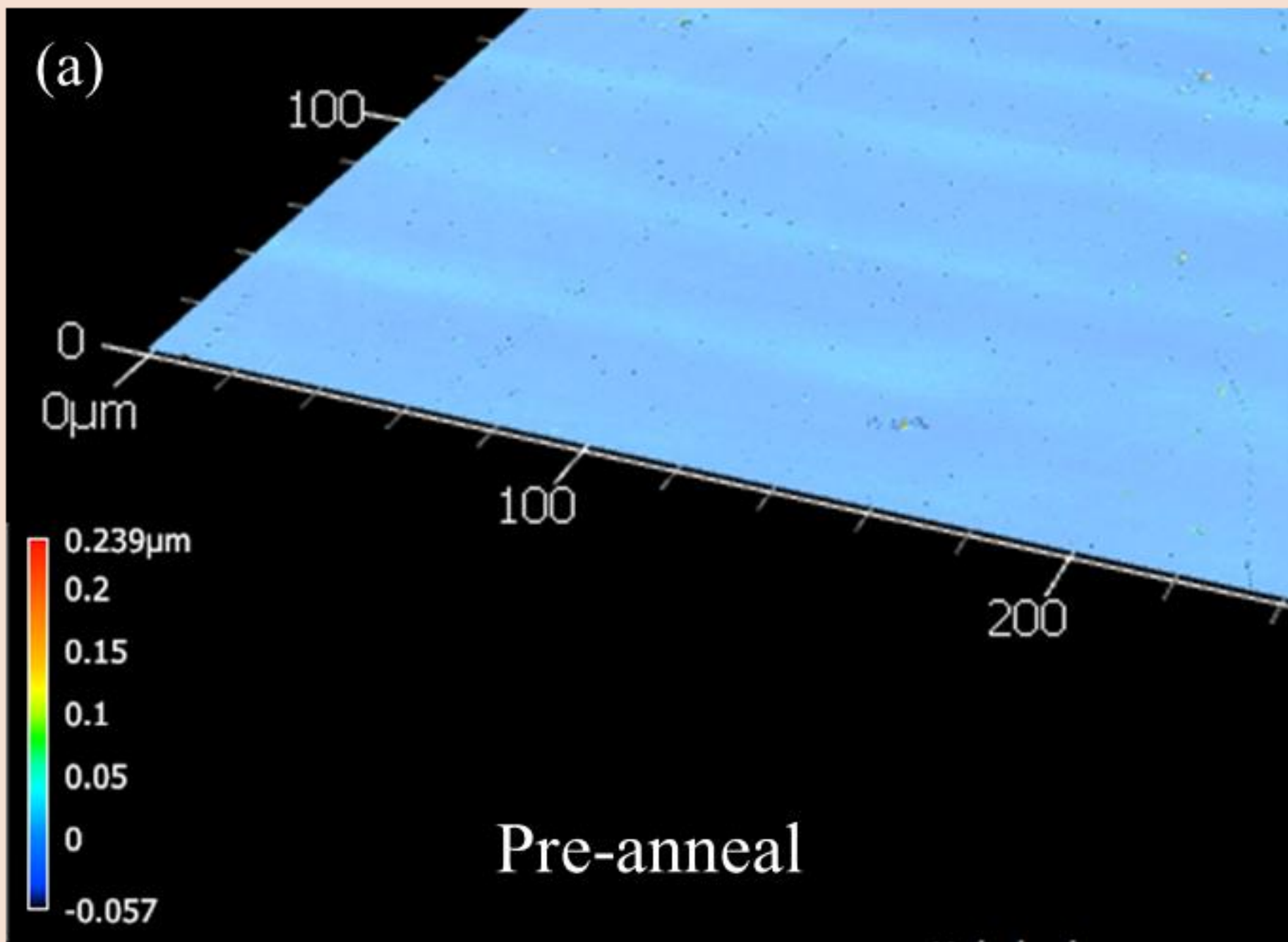
The annealed honey film shows the characteristic peaks from the honey, however there is also a strong peak at 520 cm⁻¹ from the glass substrate, because the honey film has become much thinner



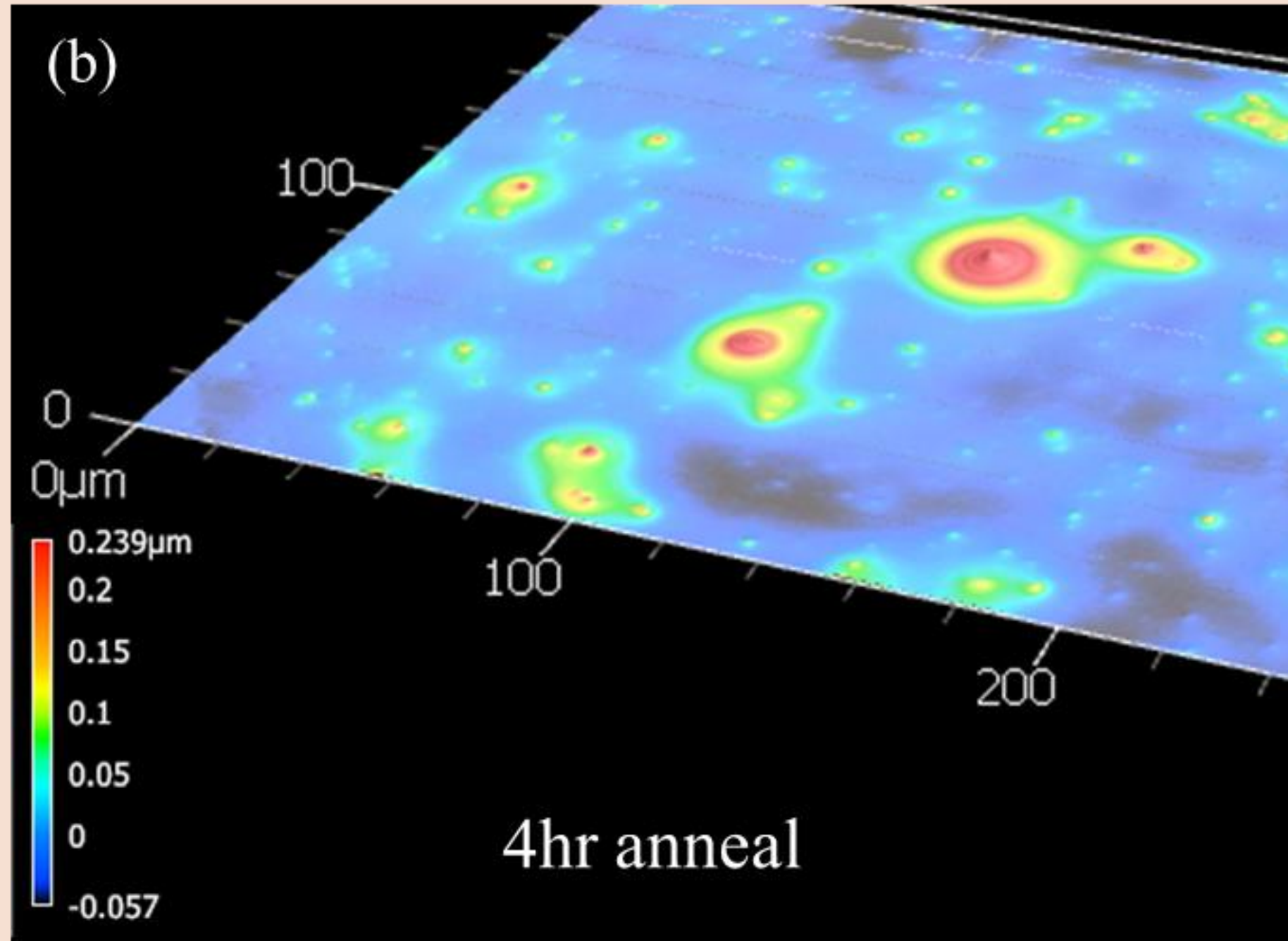
The small CNT clusters embedded in the annealed honey show peaks from the glass substrate, the honey and CNTs. The strength of the Raman signal from large CNT clusters overwhelms all others.

Laser Scanning Microscopy

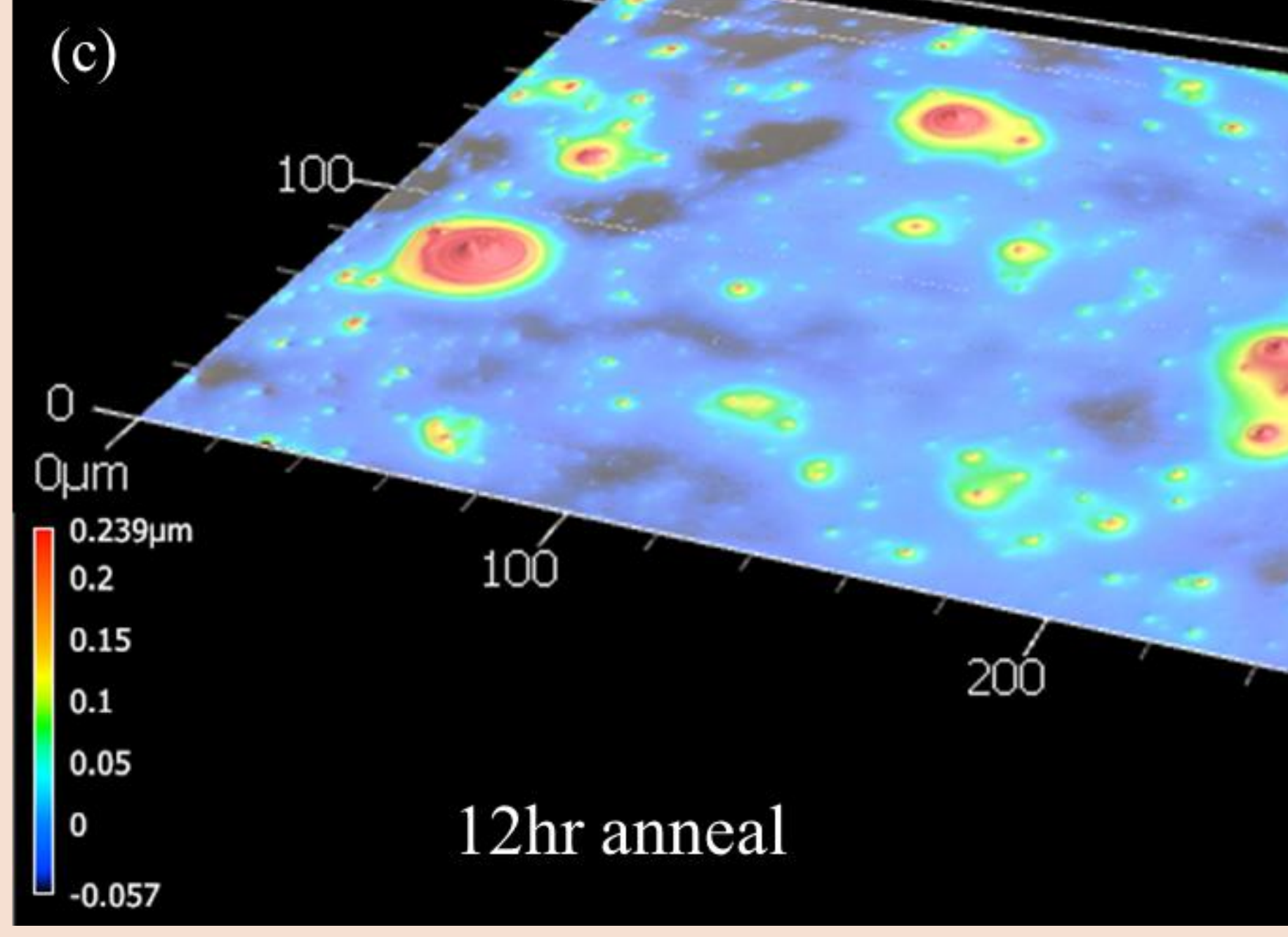
Laser scanning microscopy gives information on the surface morphology and topography of the honey film. Since the laser only scans the surface, the differences in height demonstrate how smooth the film is before and after annealing. CNT clusters are visible after annealing, and small holes in the film can be used to estimate the film thickness.



The pre-annealed film is mostly flat, with a few visible CNT clusters embedded in the film. It is estimated the film is ~1-2µm thick.



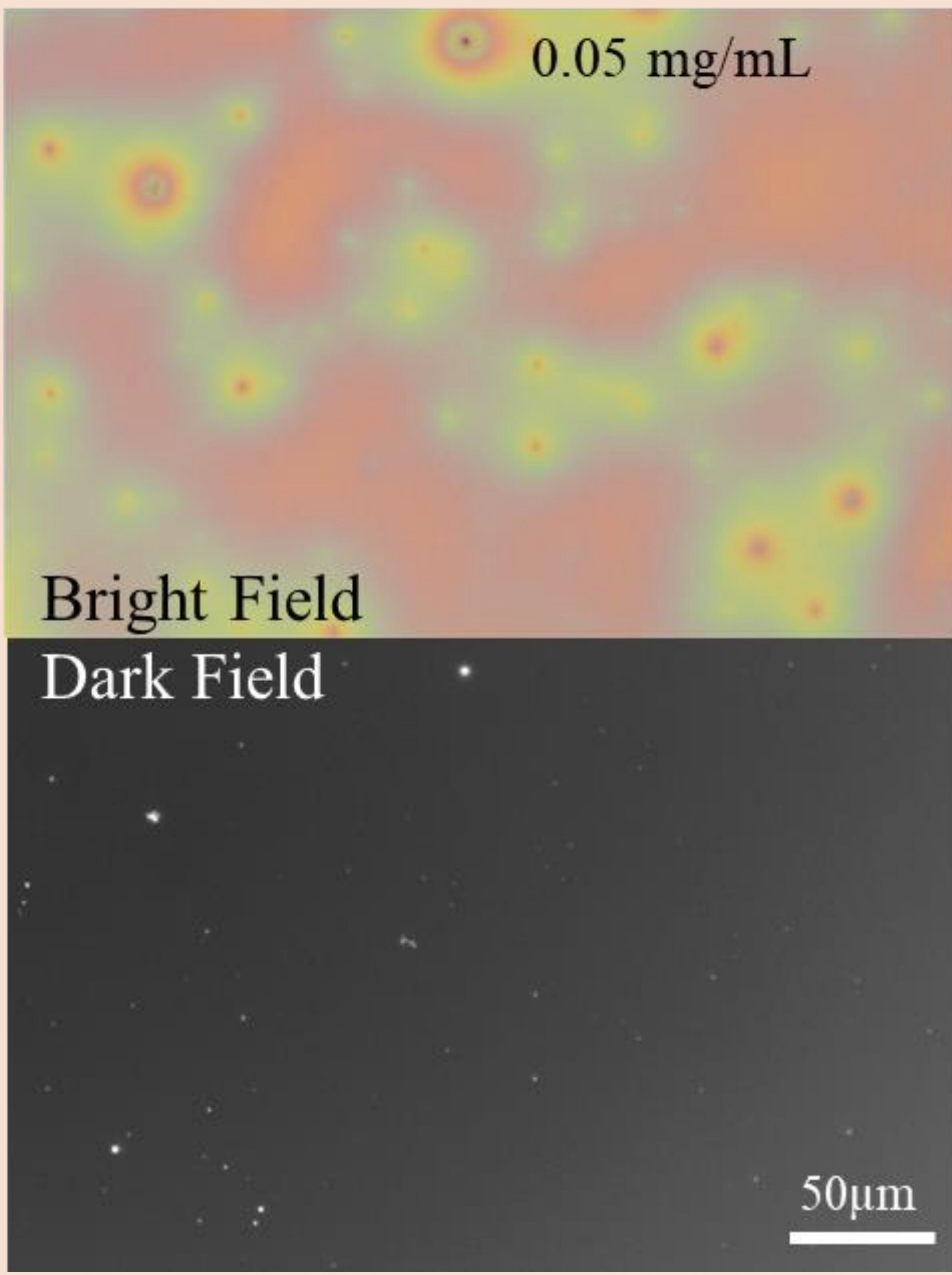
After annealing, the CNT clusters encased in honey become more visible as the film thickness decreases. After annealing the film is estimated to be 300nm thick. Small holes in the film expose the glass substrate beneath.



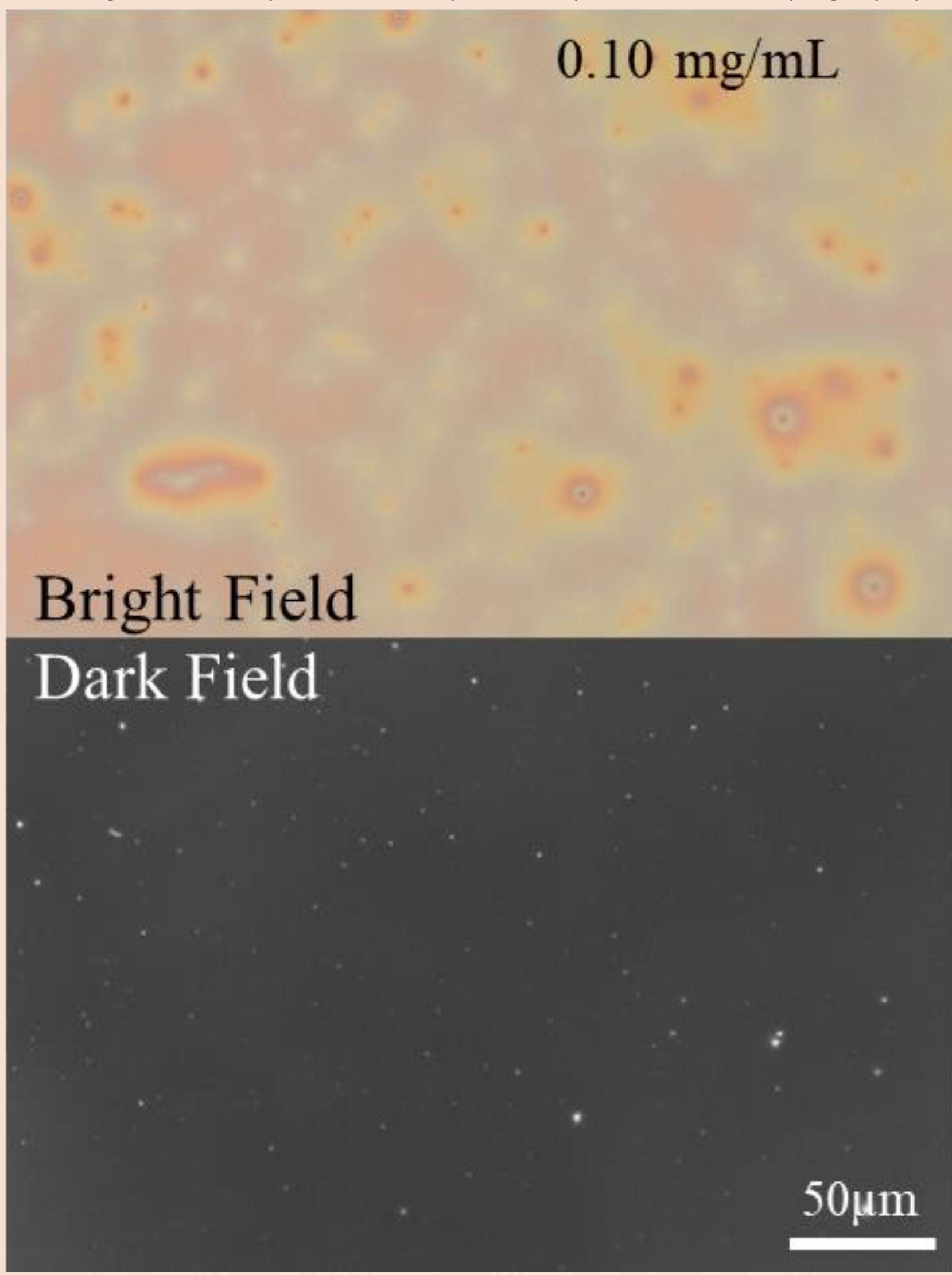
A 12 hour anneal results in the continued decrease of film thickness and the smallest CNT clusters become visible. In some areas, more holes begin to appear in the film.

Optical Microscopy

Bright field optical microscopy helps identify large CNT clusters and identify the film thickness change after annealing (large pink areas between clusters show a change to thickness). Dark field optical microscopy helps to identify individual clusters, and especially the small clusters which are not visible with bright field. Together, they can identify the dispersion and topography of the honey film with CNTs



The BF image of the honey-CNT film at low concentration shows large and small CNT clusters. The pink areas between clusters correspond to thin areas of the film from annealing. The bright points in the DF image correspond to clusters in the BF image



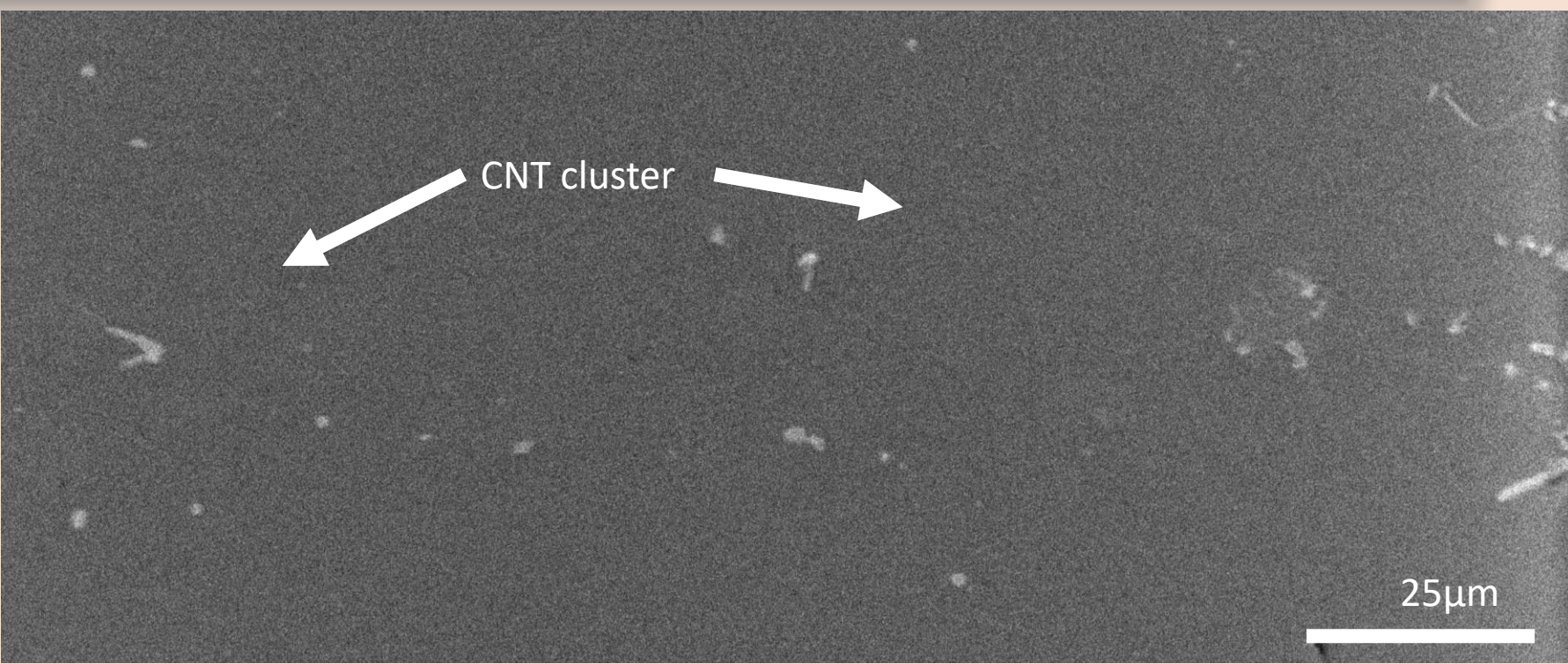
The BF image at higher concentration (0.10 mg/mL) shows fewer pink areas, indicating that the presence of CNTs in the film reduces thinning to the film.



The BF image at the highest concentration (0.20 mg/mL) shows many clusters and reduced thinning of the film. Furthermore, the dark field image demonstrates that at the highest concentration the CNTs continue to be well dispersed and do not agglomerate into larger clusters.

Concen- tration	CNT Clusters
0.05 mg/mL	~40
0.10 mg/mL	~93
0.20 mg/mL	~165

Scanning Electron Microscopy



- Using low accelerating voltage in SEM allows one to identify clusters of CNTs in the honey film.
- The clusters can be spherical groupings, or elongated bundles.
- The morphologies of the white spots are characteristic of CNT bundles. Since these spots correspond to bright points in DF optical images, confirmation with SEM allows quick future characterization with DF images.

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Conclusion & Future Work

- By developing methods to characterize honey-CNT films, a better understanding can be obtained for the role of admixture concentration and annealing time on the formation of honey-CNT films.
- This will further enable the development of memristor devices using honey-CNT films, and other applications using similar materials.
- The honey-CNT films characterized here led to a full-length manuscript published in *Advanced Materials Technologies*. <https://doi.org/10.1002/admt.202202194>
- Future work will continue to develop methods to characterize biomaterials for sustainable dielectric films